PHYC10009 Physics Foundations

Electricity and Magnetism
Week 7 Lecture 2

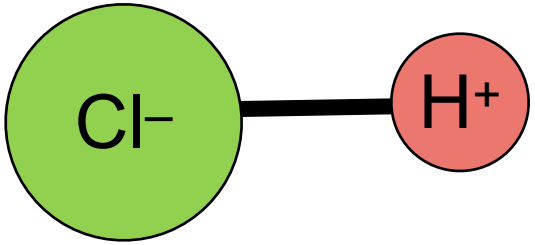
Professor Phillip Urquijo
School of Physics
University of Melbourne

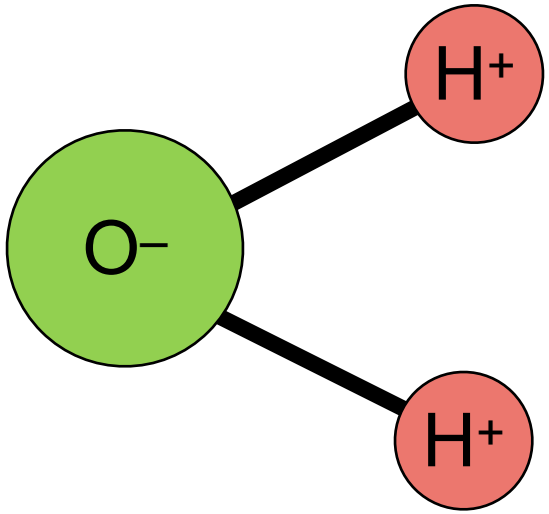
Reminder:

- Examples of using Coulomb's law $\vec{F} = k\frac{q_1 q_2}{r^2}\hat{r}$
- Concept of "field"
- Electric field $\vec{E} = \frac{\vec{F}(x, y, z)}{q}$
- Electric field around a point charge $\vec{E} = k\frac{Q}{r^2}\hat{r}$
- Example 1: field around proton $\vec{E} = 9 \times 10^9 \frac{1.6 \times 10^{-19}}{\left(0.53 \times 10^{-10}\right)^2}\hat{r} = 5.1 \times 10^{11}\text{N/C}$
- Example 2: inside He nucleus

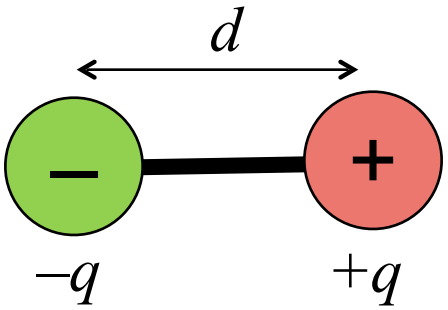
The Electric Dipole

- Electrostatic forces very strong: free charges rare in nature
- More often encounter ***DIPOLES***



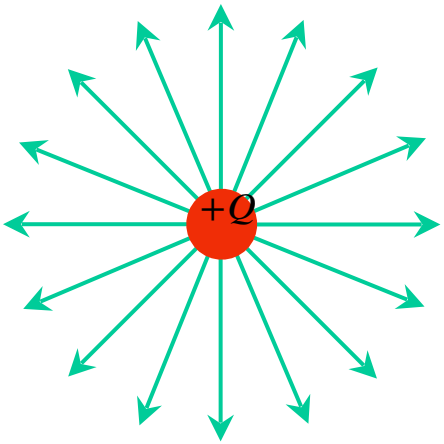


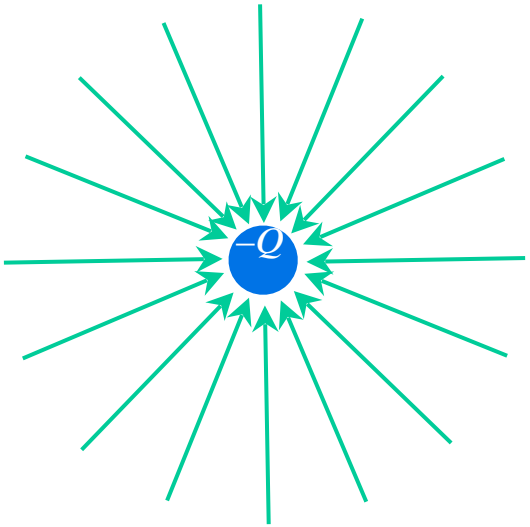
Generic dipole



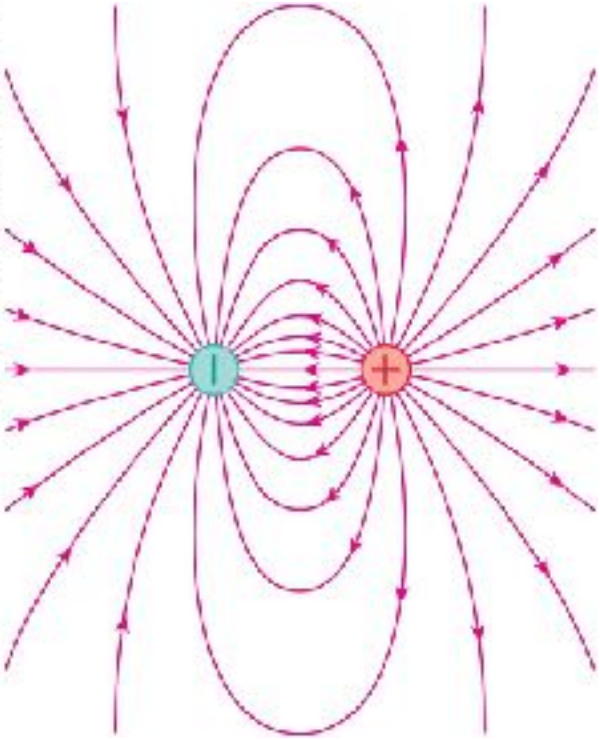
Electric field of a dipole

Field lines from a positive point source *originate* on the charge





Field lines from a negative charge *terminate* on the charge

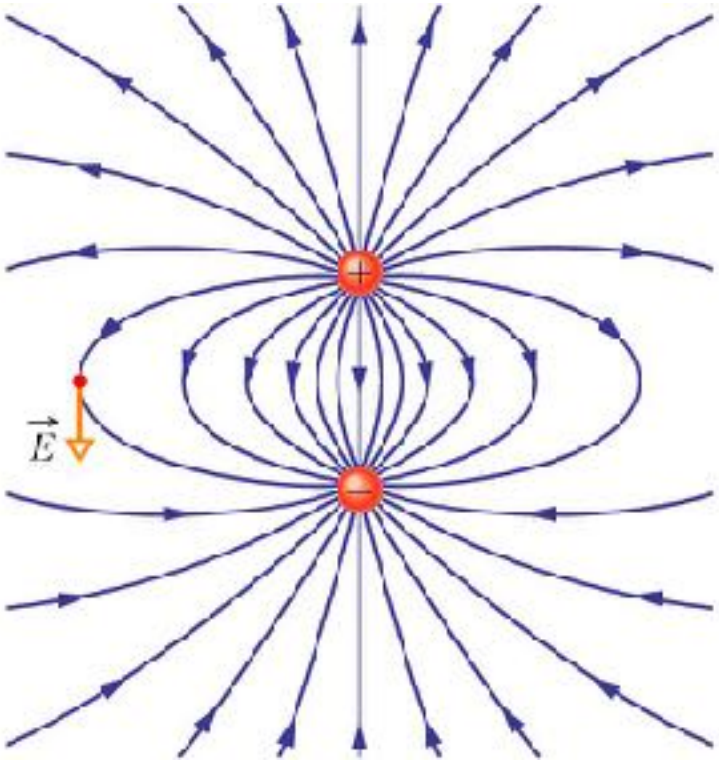


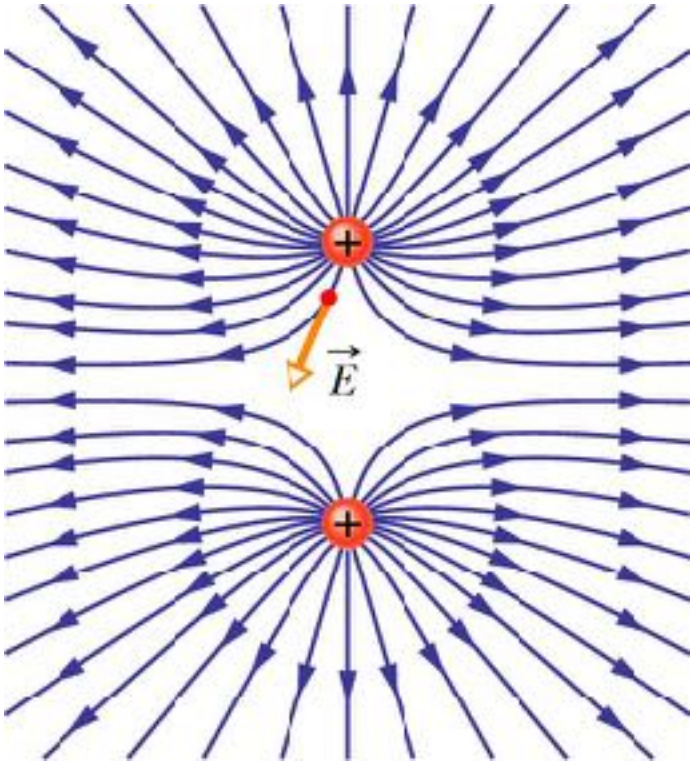
Electric field lines connect from positive to negative charges

D El-5 Visualisation of Electric Field Lines

Demo: Electric field lines

Electric Field of a Dipole

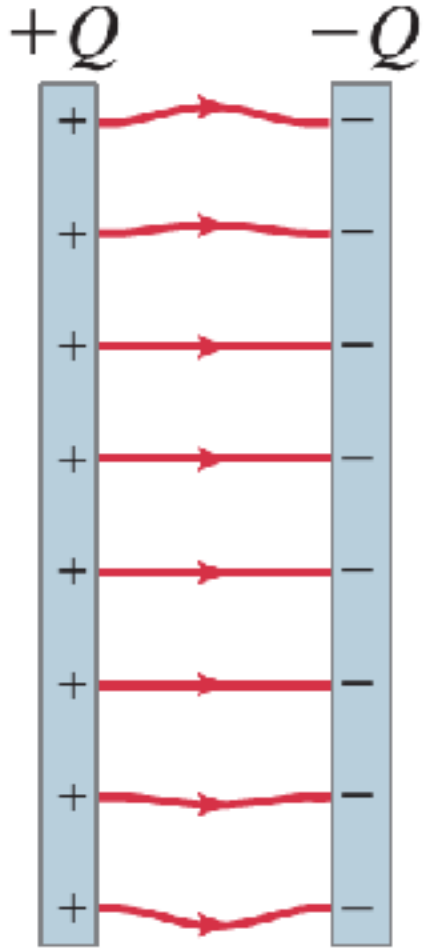




https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_en.html

Uniform electric field

A uniform collection of electric charges spread over a uniform surface creates a uniform electric field.



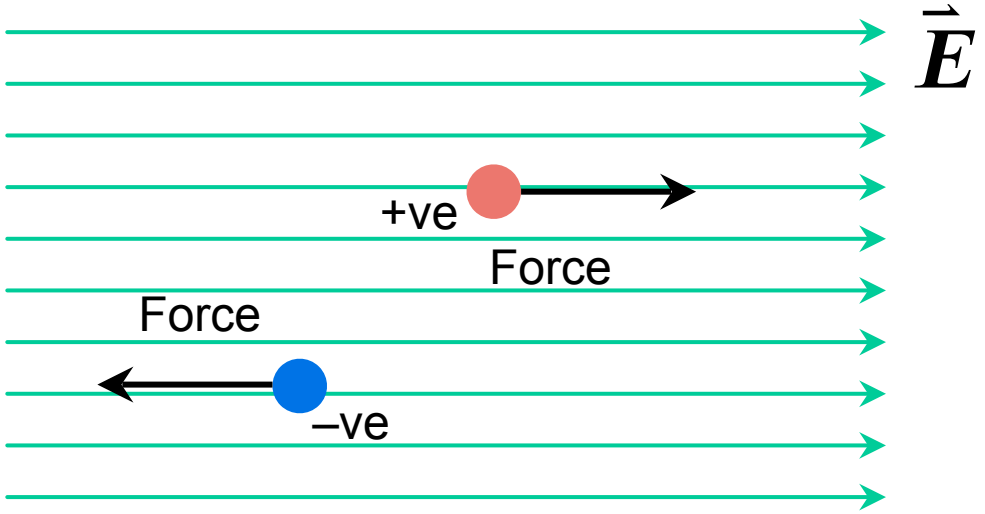
Force on charge in an electric field

If electric field is known, easy to find force on a charge in that field

(This is the *definition* of the field! Force per unit charge!)

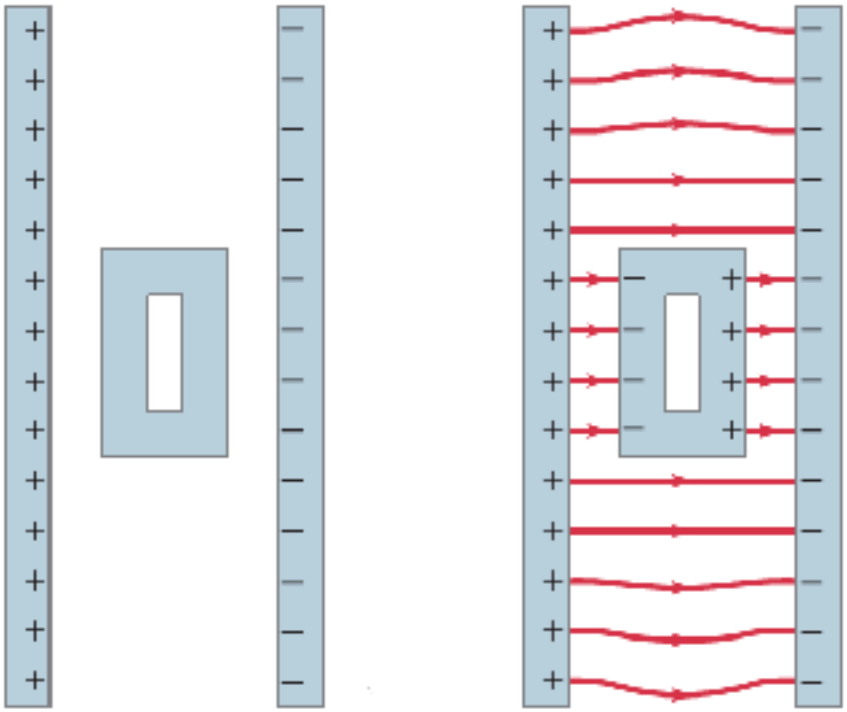
$$\vec{F} = q\vec{E}$$

- Force <u>direction</u> = (sign of charge) × (direction of field)
  - Positive charge: force in same direction as field
  - Negative charge: opposite to field



16.9 Electric fields and conductors

- $\boldsymbol{E}$ field is always normal to the surface
- $\boldsymbol{E}$ field is always zero inside a conductor
- Why?
  - Conductor: charges free to move
  - They move until field between charges balances external field
  - *Net* field zero



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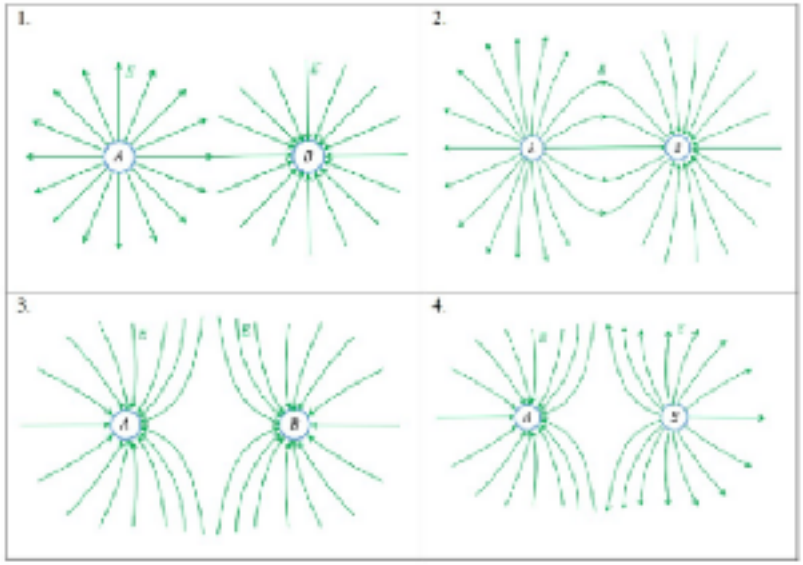
Impossible Field

Which of the 4 figures shown below represent an impossible electric field line distribution?



(1)

(2)

(3)

(4)

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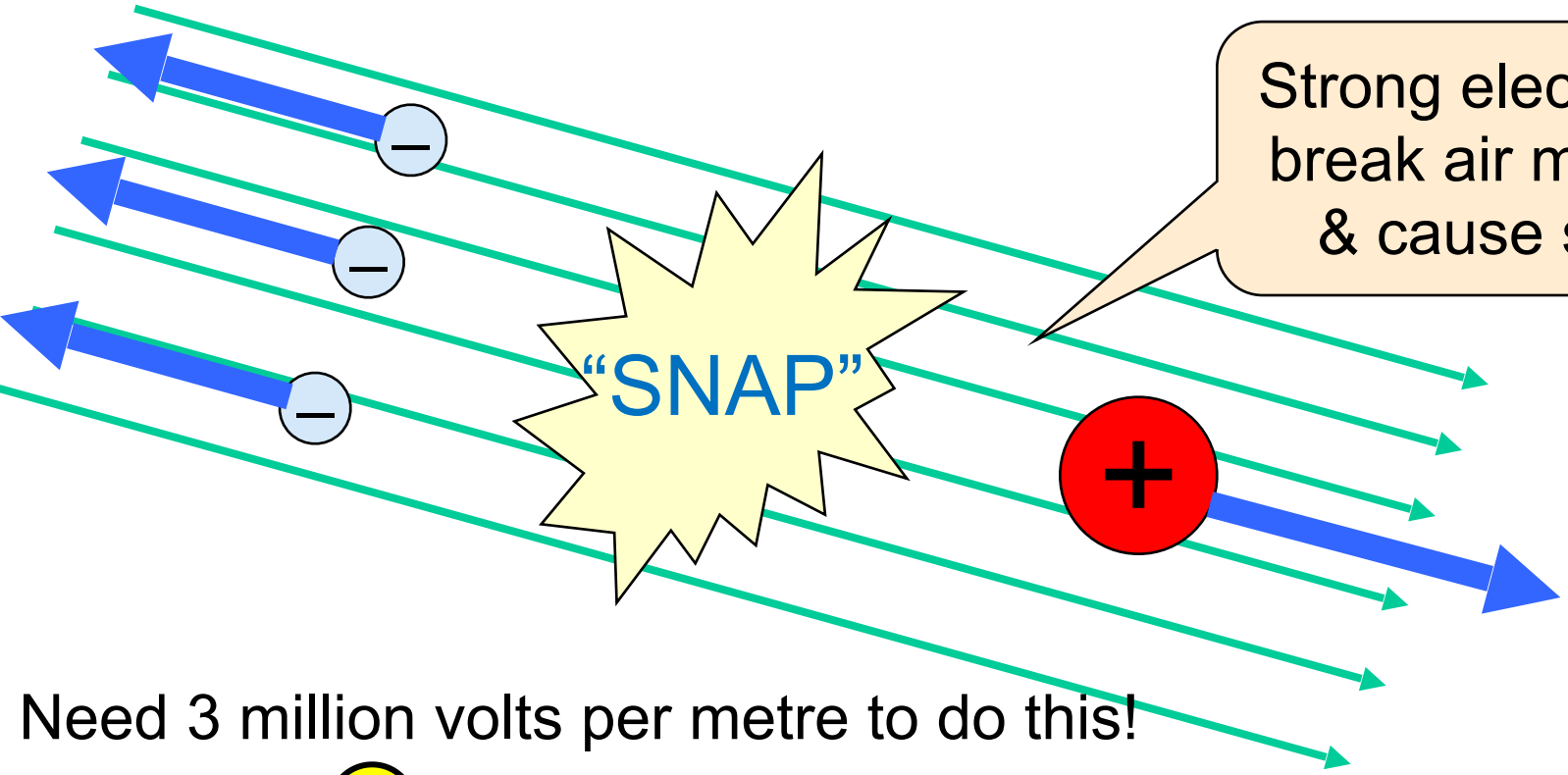
Lightning

- A molecule in a STRONG electric field



Strong electric fields break air molecules & cause sparks!

Need 3 million volts per metre to do this!

D El-8 Van de Graaff Generator (with spark)

Demo: Van de Graaf with spark

What should you do?

FIGURE 25-20 A large spark jumps to the car's body and then exits by moving across the insulated left front tire (note the flash there), leaving the person inside unharmed.

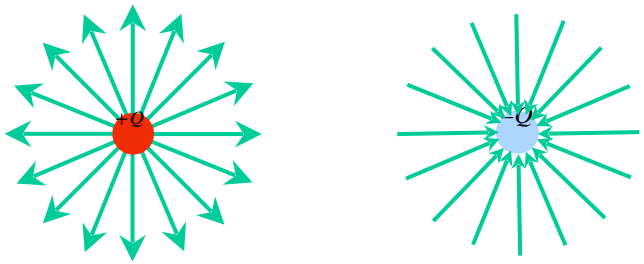
Richard Hammond hit by lightning http://www.youtube.com/watch?v=ve6XGKZxYxA&t=290s

Summary: electric field

$$\vec{E} = \frac{\vec{F}}{q}$$

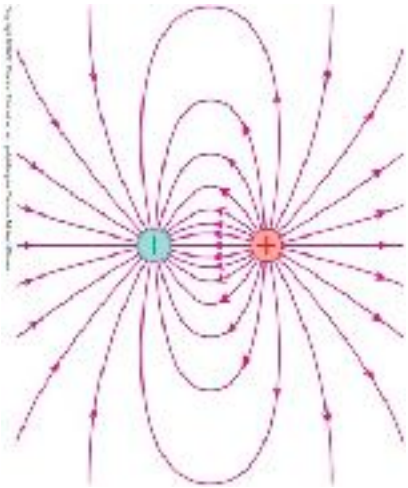
$$\vec{F} = q\vec{E}$$

Electric field of a point charge

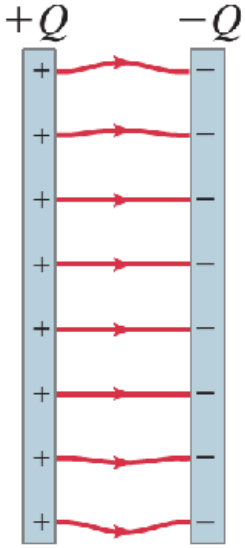


$$\vec{E} = \frac{\vec{F}}{q'} = k\frac{Q}{r^2}\hat{\boldsymbol{r}}$$

Electric field of a dipole



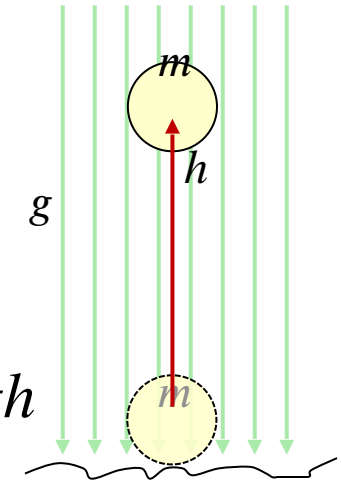
Uniform electric field

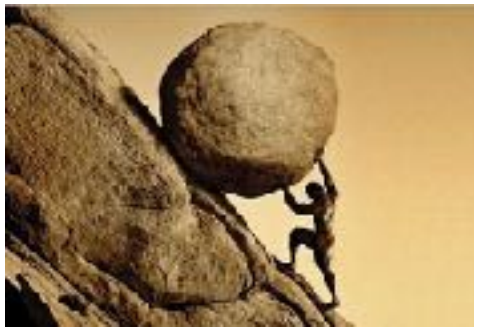


Recap: gravitational potential energy

Chapter 6

- Lift something up a hill, against gravity
  - You do work
  - Work → potential energy
  - Height of object increases





- Work: force x distance: $W = Fd = -mgh$
- Work is positive when *done by field*
  - Lift object =work *against* field = negative work

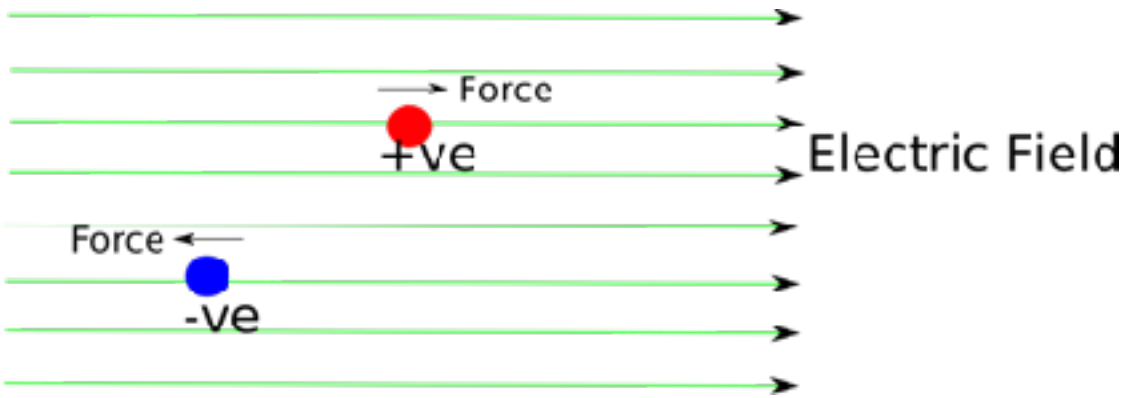
- Potential energy $\mathrm{PE_G} = mgh$
- Potential energy greater when rock higher

- Gravitational *potential* = potential energy per unit mass $\dfrac{\mathrm{PE_G}}{m} = gh$
  - Height is a like gravitational potential

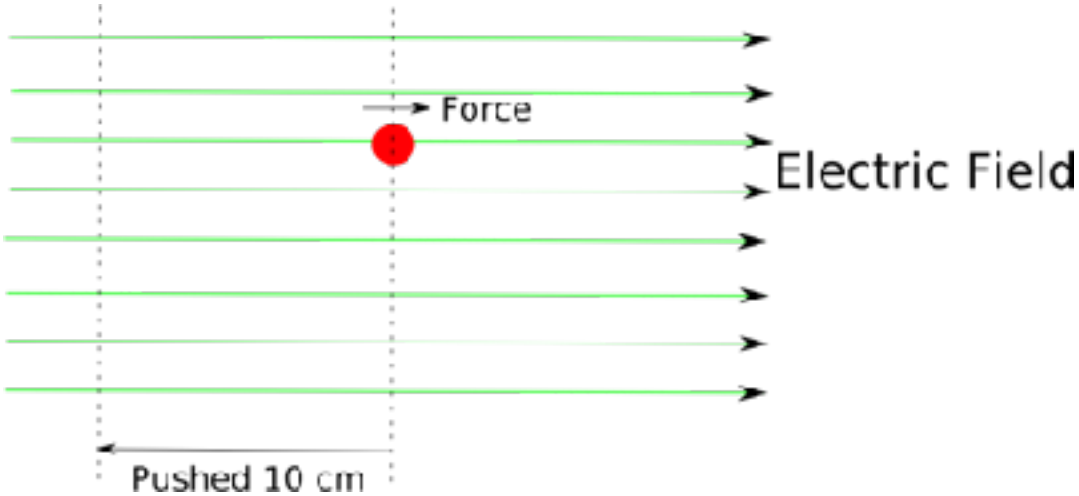
http://www.accionpreferente.com/estrategia/11-pasos-para-lograr-lo-imposible/

17.1 Work W and electric field E

Charge in electric field experiences force



Motion of a charged particle in an electric field: work required or work done



Work W and potential energy PE

Motion of a charged particle in an electric field: work required or work done?

Sign? Change in <u>potential energy</u> PE is work done *against* the field

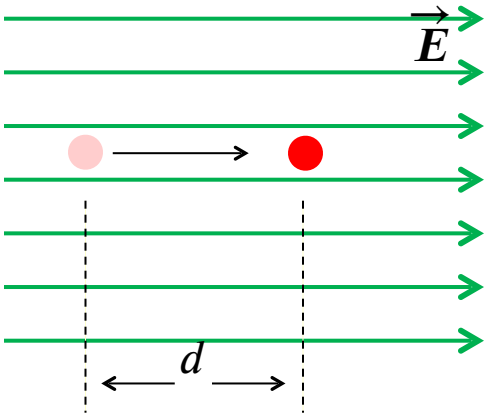
<u>Work</u> is defined in terms of work done *by the field*

- $\Delta\mathrm{PE}$ is positive if we have to push against the field
- $\Delta\mathrm{PE}$ is negative if the field moves the charge
- $W = \vec{F} \bullet \vec{d}$ thus if $\vec{F}$ and $\vec{d}$ are in the same direction W is positive

- Allow particle to move d pushed *by field*

$$W = \vec{F} \bullet \vec{d} = Fd\cos\theta = Fd > 0$$

- Work done *by* field is positive
- Potential energy reduced
- Change in potential energy is <u>negative</u>



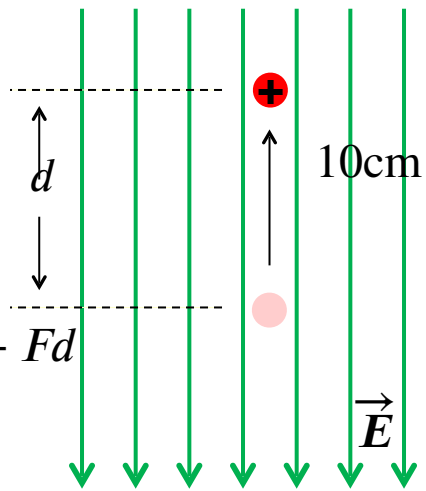
$\Delta\mathrm{PE} = -W$ where W is work done by the field

Example: Energy W and electric field E

$1\mu\text{C}$ charge, pushed 10 cm against electric field 10^5 N/C

a) How much work is done by the field?

b) What is the change in potential energy?



a) $W = \vec{F} \bullet \vec{d}$ but $\vec{F}$ and $\vec{d}$ are in opposite directions so $\vec{F} \bullet \vec{d} = -Fd$

$$W = -Fd = -qEd = -10^{-6} \times 10^5 \times 0.1 = -0.01 \text{ J}$$

This is work done *by the field*. It is negative, so in fact we have to do work to move the charge against the field.

- If released, charge moves back
  - Like lifting ball against gravity; release ball, and it falls back
- After pushing charge, it has *potential energy* PE equal to the work done *W* in moving it against the field

Potential energy

b) $\text{PE} = -W$ where *W* is the work done by the field, so $\text{PE} = Fd = +0.01 \text{ J}$

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Positive Spheres

When two positively charged small spheres are pushed together, what happens to their potential energy?

- It remains unchanged
- It decreases
- It increases
- There is no potential energy in this situation

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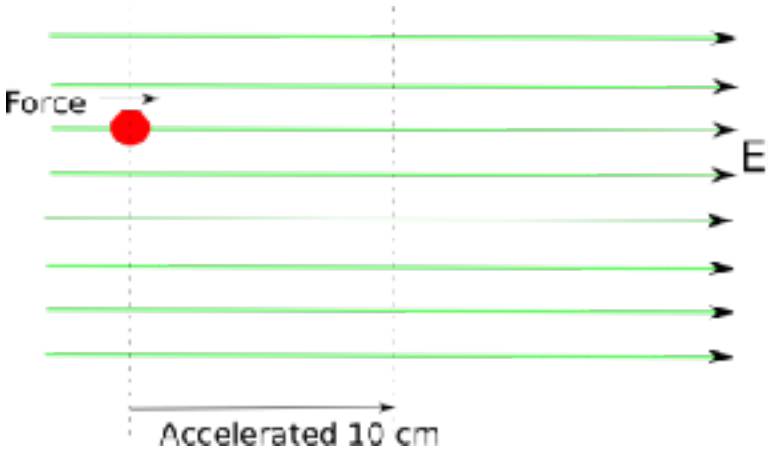
Energy W and electric field E

Consider acceleration of a charge due to force of electric field

$1\mu\mathrm{C}$ charge, mass $1\mathrm{g}$

Accelerated by electric field of 10^5 N/C

What is velocity after moving 10cm?



Change in energy: like child falling from cot

Change in kinetic energy = change in potential energy = work done by field

$\frac{1}{2}mv^2$= work = force × distance = $qEd = 0.01\ \mathrm{J}$

$$\frac{1}{2}mv^2 = 0.01 \ \rightarrow v^2 = \frac{0.02}{m} \rightarrow v = \sqrt{20} = 4.5\ \mathrm{m/s}$$

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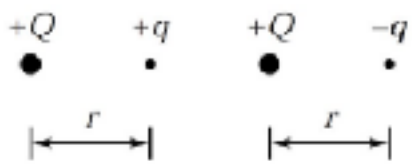
Electric potential energy

Two test charges are brought separately near a charge +Q.
First, test charge +q is brought from infinity to distance *r* from +Q.
Then +q is removed & test charge –q is brought to the same point.



The electric potential energy of which test charge is greater:

+q

-q

It is the same for both

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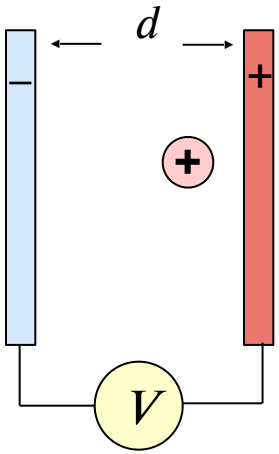
Electric potential V

Define *potential energy per unit charge:* $V = \dfrac{\mathrm{PE}}{q}$



V is the voltage or *electric potential*

Units $[V] = \dfrac{[\mathrm{PE}]}{[q]} = \dfrac{\mathrm{Joule}}{\mathrm{Coulomb}} = \dfrac{\mathrm{J}}{\mathrm{C}} = \mathrm{Volt}$

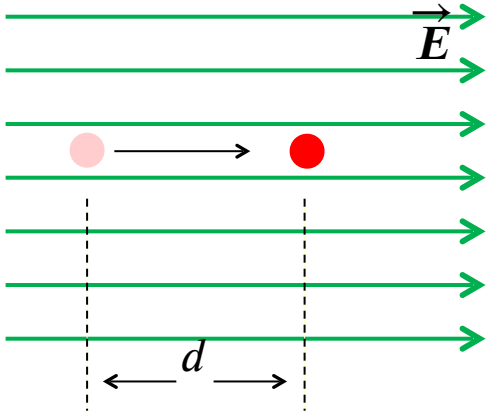
Potential energy per unit charge = electric potential ("voltage")

Confusing words: electric potential is not electric potential energy

Voltage V and field E

We know

$$V = \frac{\mathrm{PE}}{q} \text{ and } \mathrm{PE} = -qEd \text{ so } \Delta V = -Ed$$



Provided we only move along field lines:

$$\boxed{\Delta V = -Ed}$$

$$\boxed{|\Delta V| = Ed}$$

(and provided E is constant)

D El-4 The Capacitor (showing $\Delta V = -Ed$)

Summary: electric potential

- Electric potential energy $\Delta \mathrm{PE} = Fd = qEd$ (only if *E* is uniform; careful of sign)
- Electric potential energy around point charge Q $\mathrm{PE}(r) = k\dfrac{Qq}{r}$
- Electric potential = potential energy per unit charge $V = \dfrac{\mathrm{PE}}{q}$
- Electric field from potential $V = Ed$
- Potential around a point charge $V = k\dfrac{Q}{r}$
- The electron volt unit of energy $1\mathrm{eV} = 1.6 \times 10^{-19}\mathrm{J}$